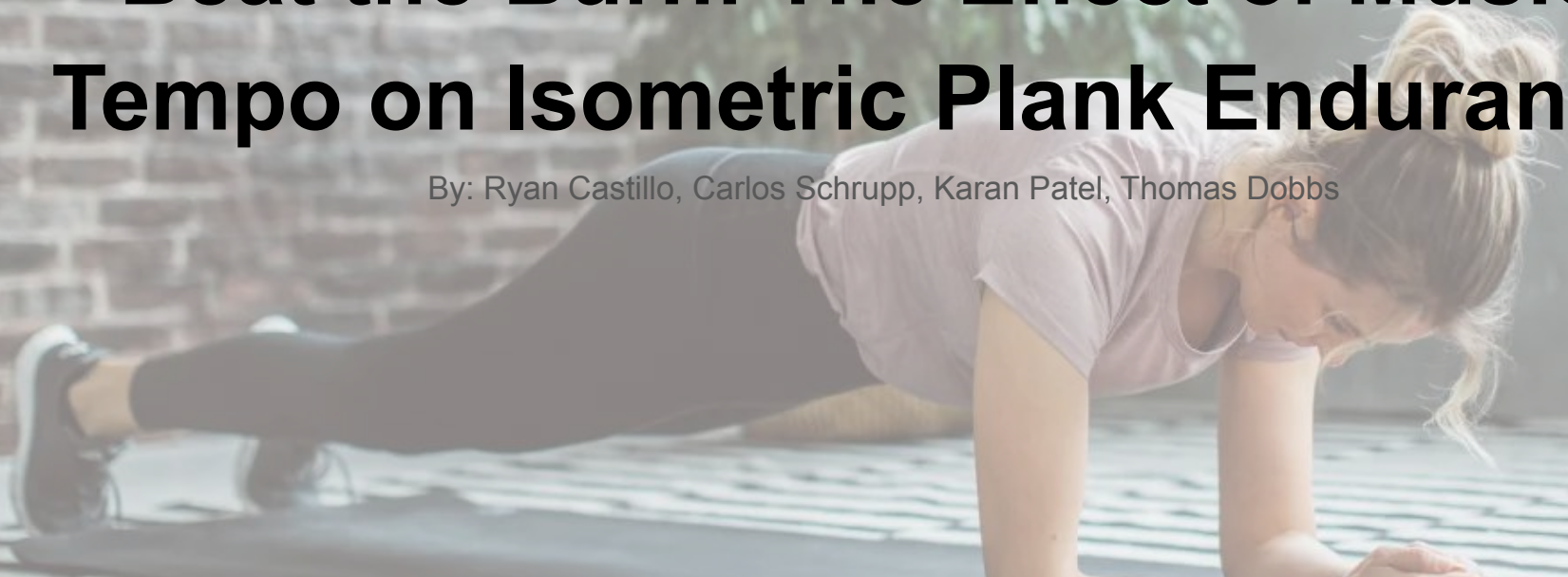


Beat the Burn: The Effect of Music Tempo on Isometric Plank Endurance

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Research Question

Does high-tempo auditory stimuli (>125 BPM) causally increase time-to-volitional-exhaustion (TTE) in a plank compared to low-tempo music (≤ 80 BPM)?

Why an Experiment?

Observational data on gym-goers is rife with selection bias (highly motivated people might just happen to like fast music). We need a randomized intervention to isolate the effect of the **beat** itself from the **person**.

Why should we care?

- **Helps us understand:** How external stimuli (like music tempo) influences physical performance, focus and fatigue.
- **Low cost performance boost:** Music is an easily accessible intervention for improving performance effectively.
- **Everyday application:** People already use music during workouts, but have little clarity on what type of music actually helps.



Hypothesis

Primary hypothesis:

Participants will hold a plank significantly longer when exposed to High-Tempo EDM.

Proposed Mechanisms of Action

Psychological Dissociation: Fast music distracts the brain from interoceptive "pain" signals (muscle burn).

Physiological Arousal: High BPM triggers the sympathetic nervous system, increasing tolerance for discomfort.

The Predicted Delta

Metric: Total Seconds Held

Direction: High-Tempo Condition > Low-Tempo Condition

Who are the units and sample?

The Participants

- **Target Population:** Adult volunteers recruited via MIDS, social media, colleagues, family & friends.
- **Sample size:**
 - 58 registered
 - 40 completed both sessions
 - 38 in the primary crossover analysis
 - 46 in the first-period benchmark (Session 1)

To ensure statistical significance and to minimize the impact of individual outliers (like that one person who can plank for 10 minutes straight).

Experimental Setting

- **Remote, Asynchronous Protocol:** Participants perform the task in their own environment. This helps test how music affects them in their real-world workout space rather than a stiff, intimidating lab.

Experiment Design

- **The Task**

Isometric Forearm Plank: Participants hold a strict form forearm plank until volitional failure (physical exhaustion).

- **The Design Type**

Randomized Within-Subjects Crossover: Every participant experiences both conditions to act as their own baseline control.

- **The Independent Variable (The Music)**

Condition A (Treatment): High-Tempo Track (≥ 125 BPM)

Condition B (Control): Low-Tempo Track (≤ 80 BPM)

- **Experimental Controls (The Constants)**

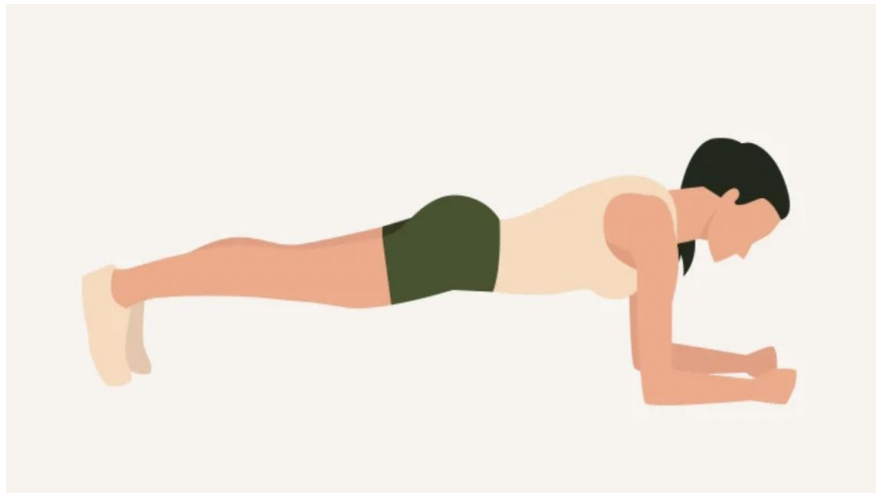
Genre: Instrumental EDM only (no lyrics to influence motivation).

Volume: Standardized to approximately 60% system volume.

Headphones encouraged / audio delivered through assigned session page

Exact fixed tracks used in each condition

Isometric Plank



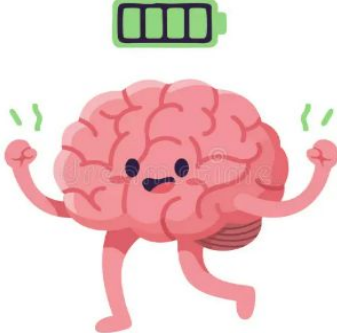
Group 1

R → High-tempo → 01: plank TTE → 24h+ washout → Low-tempo → 02: plank TTE

Group 2

R → Low-tempo → 01: plank TTE → 24h+ washout → High-tempo → 02: plank TTE

What did we measure?

Primary Outcome Variable	Secondary Outcome Variables	Pre-treatment covariates / checks
Time-to-Volitional-Exhaustion (in seconds). 	Rate of Perceived Exertion, Pre-plank self-reported Energy Level and Motivation. 	Age & Gender, General physical activity level and familiarity with planks, Use of headphones vs. speakers. 

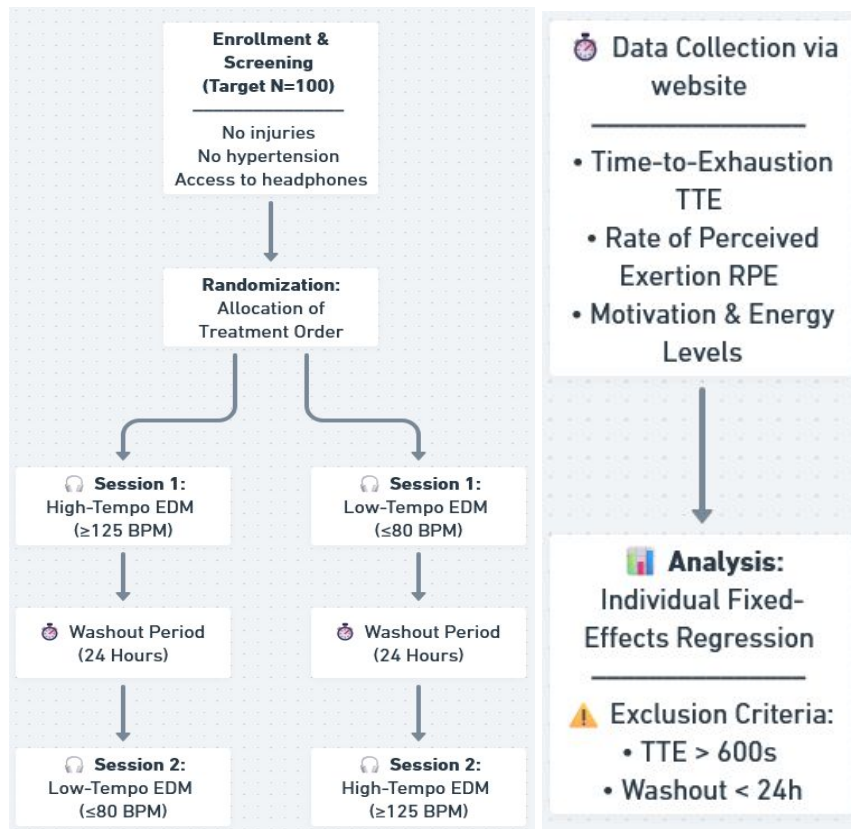
Randomization & Flow

Sequence Randomization: Participants were randomly assigned to one of two crossover pathways: **High-Tempo First** or **Low-Tempo First**.

Reason — To Mitigate Order Effects:

- **Fatigue:** Residual muscle exhaustion from the first session.
- **Learning:** Improved form or pacing strategies in the second session.
- **Carryover:** Lingering psychological motivation from the previous track.

Balance Check: Key covariates (Age, Gender, Fitness Baseline) were collected to ensure both randomized sequence groups were statistically similar and free of selection bias.



What the pilot taught us

Pilot Results

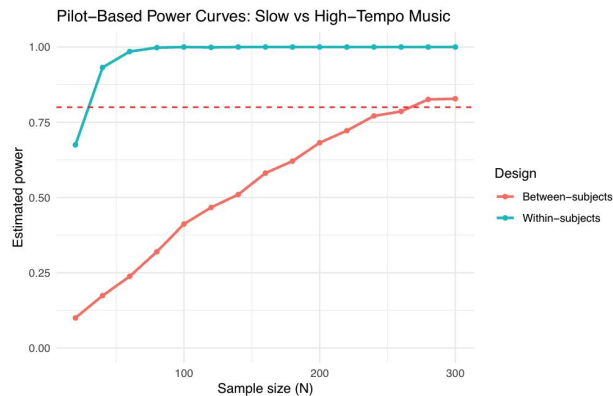
- 5 full samples (8 participants)
- High-Tempo: ~122 sec
- Low-Tempo: ~106 sec

Roadblocks [Data Quality x Friction]

- Data Quality
 - We don't know if people work out in-between
 - Stay w/in the 24+ hour window timeframe
 - Same environment for planks?
 - Participants survey accuracy (they lie)
 - Can we trust their stop-watches w/out Zoom?
 - Will people fill out the form immediately?
- Friction
 - Follow through on a 2-step x 2-day process

Table 1: Pilot-based estimates for plank time (seconds)

metric	value
Slow mean	106.20
High mean	121.80
Effect (high - slow)	15.60
SD slow	33.57
SD high	54.32
Mean diff	15.60
SD diff	27.65



How we estimated the causal effect?

The causal effect was evaluated using three progressive methods:

Model 1: Raw Paired Contrast:

A direct calculation of the mean paired difference: ($\{\text{High Tempo}\} - \{\text{Low Tempo}\}$).

Model 2: Fixed-Effects Crossover Regression:

A regression framework utilizing clustered standard errors that adjusts for period effects .

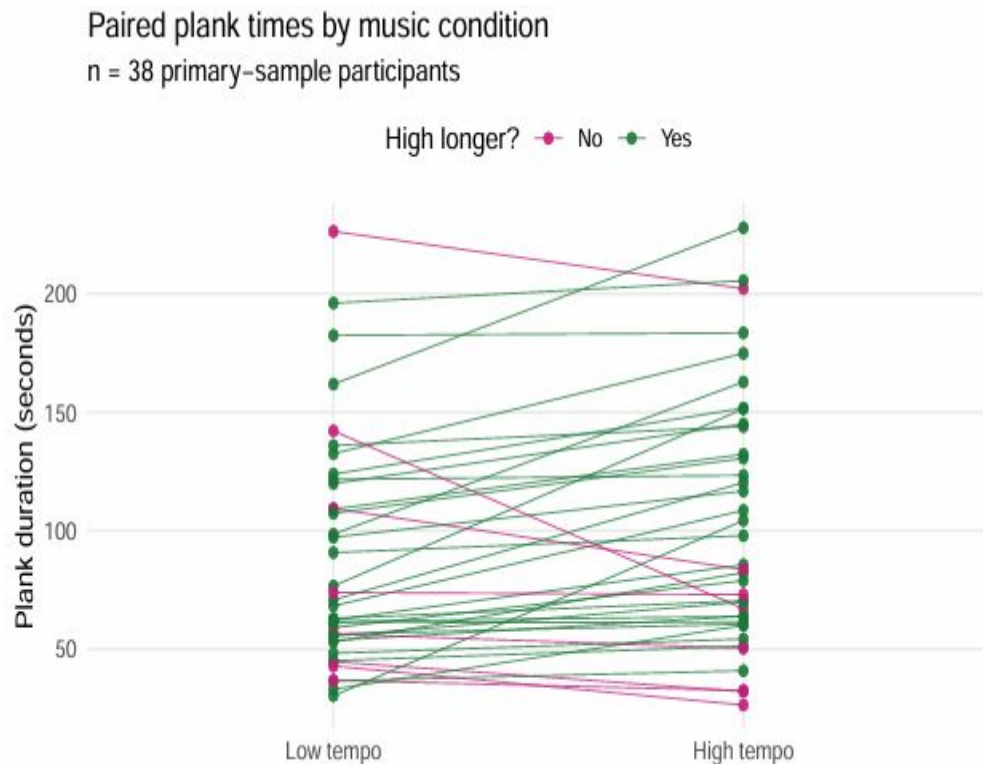
Randomization Inference (Robustness Check):

A final check simulating 10,000 permutations to test if the observed paired effect was highly unusual under the sharp null hypothesis implied by our randomization mechanism.

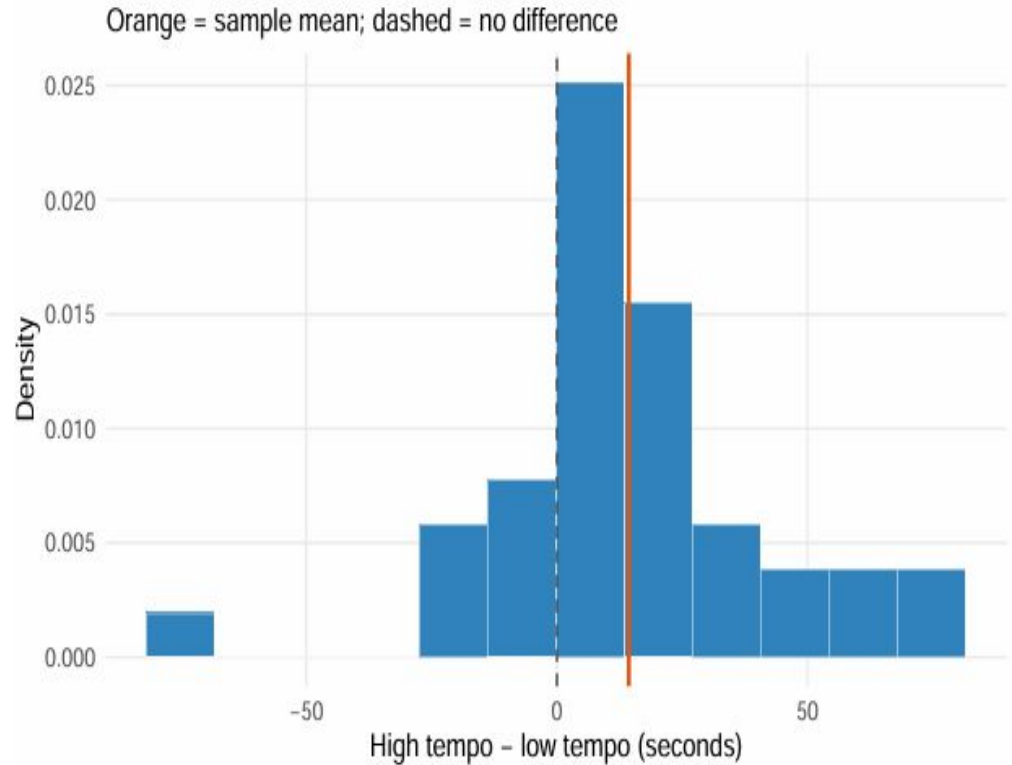
Supplementary benchmark: Session 1 only, to emulate a parallel-arm design

Did the Beat Matter?

The Statistical Verdict



Paired-Difference Distribution



quantity	value
Mean high-tempo TTE	102.4404
Mean low-tempo TTE	88.0709
Mean effect (high - low)	14.3694
95% CI lower	4.8633
95% CI upper	23.8756
Paired t statistic	3.0628
Paired p-value	0.0041
Standard error	4.6916
Paired effect size dz	0.4968

“Raw Paired Contrast” Model Results

- Participants held their planks for an average of **102.44 seconds** during the high-tempo condition, compared to **88.07 seconds** in the low-tempo control.
- On average, exposure to high-tempo music yielded a direct within-person improvement of **14.37 seconds** in total time-to-exhaustion.
- With a highly significant p-value (**p = 0.0041**), we can confidently reject the null hypothesis.
- We are 95% confident that the true average increase in endurance falls between **4.86 seconds and 23.88 seconds**.
- The paired effect size (**d_z = 0.50**) represents a "medium" effect, indicating that the music didn't just cause a statistical anomaly, but a physically meaningful boost to exercise tolerance.

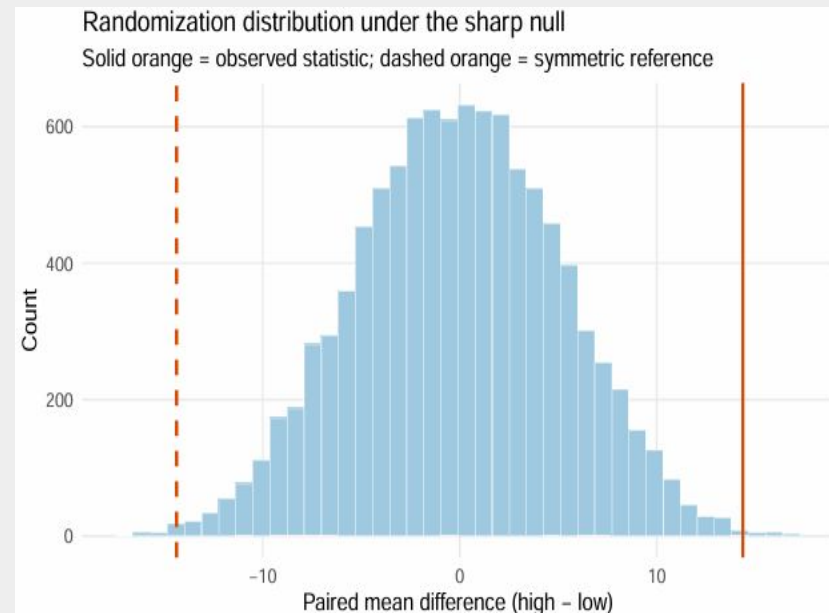
“FE Crossover Regression” Model Results

term	estimate	std_error	conf_low	conf_high	p_value
High-tempo indicator	14.3694	6.7101	1.2177	27.5212	0.0391
Period 2 indicator	3.8441	6.7101	-9.3076	16.9958	0.5703

- Even after adjusting for potential session sequence biases, the model confirms that high-tempo music increases plank endurance by an estimated 14.37 seconds.
- The high-tempo treatment effect remains statistically significant ($p = 0.039$), validating the causal claim using regression framework.
- The "Period 2 indicator" reveals a marginal average increase of 3.84 seconds during the second session, testing for minor practice, learning, or carryover effects.
- Crucially this period effect is entirely statistically insignificant ($p = 0.570$), proving that the randomized session order did not systematically bias the outcome data.
- The 95% confidence interval for the treatment effect ([1.22, 27.52]) remains strictly positive, further reinforcing the reliability of the high-tempo advantage.

Robustness Check: Randomization Inference

- **The Sharp Null Hypothesis:** Assumes the high-tempo music had absolutely zero effect on any participant's plank time.
- We computationally re-randomized the treatment assignment **10,000 times** to map all possible outcomes under the null hypothesis.
- Our observed paired difference of 14.37 seconds falls in the extreme tail of this distribution ($p = 0.0031$).
- This confirms that the treatment effect is highly statistically significant and not observed due to random chance.



Randomization Inference summary

quantity	value
Simulations	10000.0000
Observed paired mean difference	14.3694
Two-sided RI p-value	0.0031
One-sided RI p-value (high > low)	0.0014

Exploring the Mechanisms: Why Did It Work?

Paired contrast results for secondary outcomes

outcome	mean_high	mean_low	mean_diff_high_minus_low	conf_low	conf_high	p_value
RPE	7.2632	6.6842	0.5789	-0.0161	1.1740	0.0562
Motivation	5.1842	4.7105	0.4737	0.0298	0.9176	0.0371
Energy	4.6053	4.3158	0.2895	-0.1748	0.7537	0.2144

Fixed-effects treatment coefficients for secondary outcomes

outcome	estimate	std_error	p_value
RPE	0.5789	0.4237	0.1803
Motivation	0.4737	0.2905	0.1117
Energy	0.2895	0.3259	0.3803

- Participants reported a measurable bump in Motivation during the high-tempo condition (**+0.47 points**). This suggests evidence for *Physiological Arousal* theory.
- RPE increased slightly (**+0.58**). This suggests the music didn't make the exercise feel easier rather, it increased their tolerance to endure higher levels of exertion before quitting.
- No Pre-Task "Hype" (**p = 0.38**). Fast music didn't boost pre-plank energy. The psychological benefits (distraction and pain tolerance) only trigger during the physical stress.
- The motivation increase lost statistical significance in the fixed-effects model (**p = 0.11**). The psychological shifts should be treated as suggestive evidence rather than primary drivers.

- Interaction terms for both Age and Gender showed absolutely no statistical significance.
- A participant's prior physical activity level did not significantly change the treatment effect.
- To prevent false positives, we applied a strict Bonferroni correction to all exploratory subgroups.
- **The Ultimate Takeaway:** We found no Heterogeneous Treatment Effects (HTE). Therefore, the high-tempo audio stimulation is broadly applicable.

For whom did it work? (Exploratory HTE)

subgroup	interaction_term	estimate	std_error	p_value	p_value_bonferroni
Age	high_tempo:age_centered	0.2723	0.5090	0.5960	1
Gender (male vs not male)	high_tempo:male_indicator	5.5517	13.5212	0.6839	1
Activity level / fitness proxy	high_tempo:activity_level_or_0	0.8812	5.3176	0.8693	1

What this does and does not show

What This Does Show

Causal Performance Boost: High-tempo music (>125 BPM) causes a statistically significant **~14 second average increase** in plank duration.

Design Efficiency: The crossover model was nearly **3x more precise** than a standard parallel-arm comparison, successfully isolating the "music boost" from individual fitness levels.

Psychological Mechanism: High-tempo tracks significantly increase **self-reported motivation** (+0.47 points), identifying a likely driver for the endurance gain.

What This Does Not Show

Verified Precision: Because the trial was remote and self-administered, we cannot independently verify the accuracy of participant stopwatches or volume settings.

Generalizability: Findings are specific to short-duration isometric tasks (planks) and may not apply to aerobic exercise or dynamic weightlifting.

Universal Impact: No significant subgroup differences (age, gender, or fitness level), so we cannot claim a stronger effect for a specific subgroup.

Feedback we want

Measurement: Is self-reported timing acceptable given our 3x gain in precision?

- Our crossover design was nearly **3x more precise** than a parallel study (SE of 4.69s vs 13.87s).
- Does this massive gain in statistical efficiency "shield" our results from the inherent noise of unmonitored timing?

Attrition: Does our 31% dropout rate undermine the validity of the final N=38?

- We lost **20 participants** between registration and final analysis.
- We tracked this by order arm to ensure attrition wasn't biased by the music condition.
- Is our participant flow documentation enough to maintain your confidence in the "music boost"?

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